

Selecting the right compounds for screening: does Lipinski's Rule of 5 for pharmaceuticals apply to agrochemicals?

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Abstract: Large numbers of compounds are now available through combinatorial chemistry and from compound vendors to screen for lead-level agrochemical activity. The likelihood that compounds with whole-organism activity will be discovered can be increased if compounds with physicochemical parameters consistent with transport to the target site are selected for screening. Certain ranges of simple parameters (molecular mass, log *P*, hydrogen-bond donors and acceptors, rotatable bonds) have been correlated with oral bioavailability of drugs. The distribution of these parameters for commercial insecticides and post-emergence herbicides was examined and ranges consistent with whole-organism activity are proposed for the two classes of agrochemical. The most significant difference identified between drugs and these two classes of agrochemicals was the lower numbers of hydrogen-bond donors allowed in the latter cases. The frequency with which certain functional groups occur in drugs and agrochemicals was also compared.

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1 INTRODUCTION

With the advent of combinatorial chemistry in the early 1990s it became feasible for a chemist to synthesize much larger numbers of compounds for screening. In the early days of combinatorial chemistry the sheer number of compounds generated was considered virtually to guarantee that interesting leads would be discovered.^{1–4} As the field matured, the importance of the diversity of the library of compounds in improving the odds of success was recognized.^{5–9} Recently, whether the compounds generated are sufficiently appropriate for their intended end use has become a key consideration.^{10–12} Over the course of this evolution the number of compounds prepared in each library has declined considerably. During the same period large numbers of traditionally prepared screening compounds have also become available for purchase. Similar considerations of diversity and appropriateness for the intended end use can be applied to the selection of screening compounds from vendor listings. This article examines several criteria that can be applied to ensure that libraries of compounds prepared or purchased for screening as post-emergence herbicides and insecticides have appropriate characteristics for the proposed end uses.

In the pharmaceutical arena, oral dosing is the preferred delivery route. The medicinal chemist's

major challenge is to discover molecules that not only bind selectively to the appropriate receptor in man but also have the appropriate physico-chemical properties to reach that target site when delivered orally. Uptake occurs mainly in the small intestine *via* the intestinal epithelium.¹³ A further barrier to uptake of central nervous system drugs is the blood/brain barrier which tends to be more difficult to cross than the intestinal epithelium.^{14–16} Numerous studies have been published which attempt to correlate oral bioavailability with measured or calculated physical properties of compounds.^{17–21} In addition, sophisticated approaches to distinguishing drug-like from non-drug-like molecules have been published.^{22–24} In a recent influential publication Lipinski *et al*¹¹ laid out a set of empirically derived rules, 'the Rule of 5', that delineate the physico-chemical properties and molecular features of orally bioavailable pharmaceuticals. In their study, the structures of over 2000 pharmaceuticals that have entered Phase II clinical trials were analyzed in terms of molecular mass, calculated log *P*, number of hydrogen-bond donors and number of hydrogen-bond acceptors. Log *P* is the logarithm of the octanol/water partition coefficient, and two different methods of calculation were employed, Hansch's clog *P*²⁵ and Moriguchi's mlog *P*.²⁶ The results showed that orally bioavailable pharmaceuticals typically have molecular mass ≤ 500, clog *P* ≤ 5 (mlog *P* ≤ 4.15), number of

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H-bond donors ≤ 5 and number of H-bond acceptors ≤ 10 . The Rule of 5 has the advantage of simplicity and a readily understood physico-chemical basis. Furthermore, computer algorithms can readily be implemented to calculate all the parameters employed in the Rule of 5 for large numbers of compounds. By contrast, many of the other approaches to the prediction of oral bioavailability and other transport processes in biological systems are substantially more computationally intensive or use parameters that are not readily available.^{14–21} The Rule of 5 has been widely applied to refine the selection of compounds and libraries of compounds to be purchased for random screening, and to guide the medicinal chemist towards the synthesis of compounds with a higher probability of the latter becoming drugs. Not surprisingly, given its simplicity, the Rule of 5 does not always correctly distinguish compounds that are and are not orally bioavailable.²⁷ Taking a similar approach to Lipinski's, Ghose *et al.*¹² have reported on an examination of more than 6000 pharmaceuticals in the Comprehensive Medicinal Chemistry (CMC) database in terms of *alog P*, molar refractivity, molecular mass, total number of atoms and frequency of occurrence of 24 different functional groups.¹² The *alog P* is a method for calculating *log P* developed by Ghose and Crippen.^{28–30} Ghose *et al.* defined the following qualifying ranges that include 80% of the compounds studied: molecular mass = 160–480, total number of atoms = 20–70, molar refractivity = 40–130, *alog P* = –0.4 to 5.6. Their results for molecular mass and *alog P* are very comparable to Lipinski's results with molecular mass and *clog P*. They also examined the same set of parameters for certain subsets of drugs such as anti-psychotics and anti-neoplastics and showed that these can occupy different qualifying ranges.

Spraying is the preferred route of delivery for most crop-protection chemicals. The central challenge for the agricultural chemist is to discover molecules that bind selectively to the appropriate receptor in the target species and also have the appropriate physico-chemical properties to reach that target when sprayed in the field. It is not unreasonable to expect that the constraints of the physical properties of a molecule to express activity in the field may be different from those for oral bioavailability. Certainly, agrochemicals are applied less frequently than pharmaceuticals are dosed. However, there is greater scope to adjust the formulation of a crop-protection chemical and to use adjuvants such as surfactants, oils, fertilizers and pH modifiers to optimize penetration.^{31–34} The barriers to uptake which crop-protection chemicals must cross vary. Post-emergence herbicides must generally be able to penetrate the leaf cuticle^{35,36} to have whole-plant activity, and often have superior properties when they are phloem-mobile.^{37–42} Uptake through the leaf cuticle is somewhat analogous to trans-dermal delivery of drugs,⁴³ since both the leaf cuticle and the skin mainly protect the organism from the external

environment by preventing water loss and excluding exogenous chemicals. In contrast root and shoot uptake and xylem mobility are more important for pre-emergence herbicides.^{44,45} The industry-wide observation that pre-emergence herbicide leads are discovered more often than post-emergence leads may in part reflect that roots are the major plant organs involved in uptake of nutrients and that the criteria for xylem mobility are less stringent than those for phloem mobility. The situation for insecticides is more complex than for herbicides because of the interactions between the plant and the insect. For contact activity, insecticides must be absorbed through the insect integument; however, insecticides may be taken up by the plant which is eaten by the insect, in which case the compound may be absorbed into the insect through the midgut.^{46–49} No attempt has been made in this study to distinguish between insecticides that act primarily through contact activity and primarily as stomach poisons. Fungicides are not considered in the current study. Despite the differences in barriers to uptake, the requirement for the compounds to cross membranes is universal.

It was of interest to determine to what extent crop-protection chemicals conform to the Rule of 5, or whether these constraints should be adjusted when purchasing compounds or designing libraries for random screening for agrochemical utility. To this end, parameters similar to those examined by Lipinski *et al.*¹¹ and Ghose *et al.*¹² were generated for 136 post-emergence herbicides and 243 insecticides.

2 EXPERIMENTAL METHODS

Sets of commercial and late development post-emergence herbicides (136 compounds, Table 1) and insecticides, miticides and acaricides (243 compounds, Table 2) were identified from published sources.^{50–53} All acidic compounds were considered in their unionized form, and salts were considered in their protonated form. Moriguchi *log P* values (*mlog P*)²⁶ were calculated in Sybyl 6.5⁵⁴ using an in-house modification of the SPL program described by Blake.⁵⁵ The numbers of hydrogen-bond donors were obtained by counting the numbers of OH and NH bonds in each molecule, as described by Lipinski.¹¹ The numbers of hydrogen-bond acceptors were approximated by summing the numbers of nitrogen and oxygen atoms, again as described by Lipinski *et al.*¹¹ *alog P* values^{28–30} and numbers of rotatable bonds were obtained from the Cerius2 Version 3.8 QSAR module.⁵⁶ Acidic compounds were identified by inspection. Functional group frequencies were obtained by sub-structure searches as described by Ghose *et al.*¹²

3 RESULTS AND DISCUSSION

The distributions of molecular masses for the post-emergence herbicide and insecticide sets of com-

pounds are shown graphically in Fig 1. In both cases the 300 to 400 molecular mass range is the most populated bin. The average molecular masses are 329 for the herbicides and 324 for the insecticides. It is readily apparent that the Rule of 5 molecular mass limit < 500 includes the majority of both classes of agrochemical. Increasing molecular mass or size has frequently been observed to be correlated with lower solubility and poorer penetration through membranes.^{10,11,13}

The distributions of two different calculated log P values, mlog P and alog P , for the two sets of compounds are shown in Figs 2 and 3. The mlog P and alog P procedures are both robust atom-based methods of calculating log P which produce a value for almost any structure; however, the values are often less accurate than those from fragment-based methods such as clog P .^{11,25,57} The drawback of the more accurate fragment-based methods is that they often fail due to missing fragment values, which limits their general utility, and hence they were not examined in this study. Values of mlog P for herbicides are centred in the 1.2 to 2.4 range, with an average of 1.7, while insecticides are centered in the 1.7 to 3.5 range with an average of 2.6. Values of alog P for both classes exhibit a much broader distribution than those for mlog P , with the herbicide alog P distribution centered in the 1.4 to 3.6 range and that for insecticides in the 2.5 to 4.8 range. The average alog P values for both classes are approximately one log unit higher than the corresponding average mlog P values. Despite the numerical differences in the values given by the mlog P and alog P calculations, the insecticide set shows a clear tendency towards higher values of both mlog P and alog P than do the herbicides. The Rule of 5 limit of 4.15 for mlog P seems appropriate for herbicides but would exclude many insecticides. It is generally recognized that compounds with log P values in the range -0.5 to 3 are optimal for crossing membranes, and that compounds with log P values above this range tend to accumulate in lipid membranes, while very hydrophilic compounds cannot partition into the membrane.¹³ Nonetheless many insecticides have log P values substantially higher than 3 . The relevance of water/octanol partition to sprayed agrochemicals may be questioned. When the spray droplet is first deposited on the leaf surface, the agrochemical is in a largely aqueous environment from which it must partition into the more lipophilic leaf cuticle. However, water quite rapidly evaporates from the spray droplet, leaving on the leaf the agrochemical and various low-volatility formulation components which undoubtedly significantly influence partition behavior. In addition many herbicides are weak acids (*vide infra*) and their partition behavior depends on the pH of the medium.

The distributions of hydrogen bond donors for both herbicides and insecticides (Fig 4) are skewed towards low values, with 93% of herbicides having two or fewer hydrogen-bond donors, while a full 73% of the

insecticides have no hydrogen-bond donors at all. Overall, a significant adjustment of the Rule of 5 limit for hydrogen-bond donors seems warranted for both types of crop-protection chemical considered here. Compounds with large numbers of hydrogen-bond donors and acceptors are often rather insoluble in many solvents, due to the formation of intermolecular hydrogen bonds in the crystal lattice.

Herbicides exhibit a broad distribution in numbers of hydrogen-bond acceptors (Fig 5a). As with donors, numbers of hydrogen-bond acceptors in the insecticide set are skewed towards lower values than in herbicides (Fig 5b). Some of the older insecticides, eg DDT, lindane have no oxygen or nitrogen atoms. In addition to the upper limit used by Lipinski, lower limits for hydrogen-bond acceptors may be warranted, at least for herbicides.

Although not discussed in the Rule of 5, the number of rotatable bonds in a compound is often considered to be an important determinant of biological activity. Examination of Fig 6 suggests that the upper limit on number of rotatable bonds should be set around 12. No difference in the distribution of this parameter between herbicides and insecticides is discernible. Compounds with large numbers of rotatable bonds have poor ΔG of binding to their target sites because of large ΔS terms engendered by the need to freeze many rotatable bonds upon receptor binding. Thus very flexible compounds are rarely potent drugs or pesticides. It should be noted that, in this analysis, all single bonds are considered to be rotatable although, in fact, many are not freely rotating; the amide C-N bond is the most obvious example and a fairly common element in the structures considered here (Table 3).

Table 3 shows the frequency of occurrence of a number of functional groups in the herbicide and insecticide data sets, and compares this with their frequency of occurrence in the CMC database as reported by Ghose *et al.*¹² Experienced agricultural chemists recognize the paucity among crop-protection chemicals of certain functional groups that are common in pharmaceuticals. In particular, alcohols and amines are among groups that are much less common in agrochemicals than in pharmaceuticals. The low frequency with which alcohols, primary amines and secondary amines occur among agrochemicals may be due to their relatively easy metabolism by conjugation with sugars or by oxidation.⁵⁸ Crop-protection chemicals must generally persist in the field for days or weeks to be of practical value. The *in vivo* half-lives of pharmaceuticals, however, generally fall in the 12–30 h range. Heterocyclic rings are approximately equally prevalent in pharmaceuticals and herbicides, but less so in insecticides; however, aromatic heterocycles make up the majority of heterocycles found in herbicides. Aromatic rings are likely to be more stable in the environment than alicyclic rings, and might be expected to be more prevalent among agrochemicals than pharmaceuticals. Such a trend is apparent in the heterocyclic rings in herbicides, most

Table 1. Post-emergence herbicide training set

Name	Molecular mass	mlog P	alog P	H-bond donors	H-bond acceptors	Rotatable bonds
2,4-D	221.0	1.16	2.37	1	3	4
2,4-DB	249.1	1.73	2.80	1	3	6
Acifluorfen	361.7	2.46	4.53	1	6	5
Alloxydim	323.4	1.37	4.53	1	6	8
Amidosulfuron	369.4	-0.17	-1.60	2	12	8
Amitrole	84.1	-0.09	-0.11	2	4	1
Anilofos	367.9	1.78	3.76	0	4	8
Asulam	230.2	-0.78	0.03	3	6	5
Azafenidin	338.2	3.69	3.29	0	5	4
Azimsulfuron	424.4	1.15	-0.42	2	15	8
Benazolin	243.7	1.08	1.81	1	4	3
Benfuresate	256.3	0.84	1.82	0	4	3
Bensulfuron-methyl	410.4	1.51	0.86	2	11	10
Bentazone	240.3	0.61	0.68	1	5	1
Benzofenap	431.3	3.32	4.89	0	5	6
Bilanofos	323.3	-1.96	-4.68	6	9	13
Bispyribac	430.4	2.72	3.48	1	12	10
Bromofenoxim	461.0	2.27	5.30	1	9	6
Bromoxynil	276.9	1.18	3.38	1	2	2
Butroxydim	399.5	2.74	5.44	1	5	9
Cafenstrole	350.4	3.38	3.33	0	7	6
Carbetamide	236.3	1.27	1.32	2	5	7
Carfentrazone-ethyl	412.2	4.08	4.18	0	6	7
Chlorimuron-ethyl	414.8	1.42	1.80	2	10	9
Chloroxuron	290.8	2.53	3.06	1	4	5
Chlorsulfuron	357.8	2.27	1.93	2	9	6
Cinosulfuron	413.4	1.43	0.50	2	12	11
Clethodim	359.9	2.45	3.96	1	4	10
Clodinafop	311.7	1.62	3.35	1	5	6
Clopyralid	192.0	0.54	2.24	1	3	2
Cloransulam-methyl	429.8	1.54	2.82	1	10	7
Cyclosulfamuron	421.4	2.16	0.90	3	11	10
Cycloxydim	325.5	2.30	3.16	1	4	7
Cyhalofop-butyl	357.4	1.90	4.72	0	5	10
Dalapon	143.0	1.00	1.53	1	2	2
Desmedipham	300.3	2.52	3.02	2	6	8
Dicamba	221.0	1.16	2.53	1	3	3
Dichlorprop	235.1	1.45	2.91	1	3	4
Diclofop-methyl	341.2	2.20	4.37	0	4	6
Diclosulam	406.2	2.03	3.61	1	8	5
Difenzoquat	249.3	3.53	3.80	0	2	2
Diflufenzopyr	334.3	1.68	1.87	3	7	7
Dimefuron	338.8	3.54	3.54	1	7	5
Dinoterb	240.2	1.71	3.30	1	7	4
Diquat	184.2	1.59	2.16	0	2	0
Diuron	233.1	2.26	2.15	1	3	3
Ethametsulfuron-methyl	410.4	2.16	0.82	3	12	10
Ethidimuron	264.3	0.24	0.01	1	7	5
Ethofumesate	286.3	1.13	1.87	0	5	4
Ethoxyfen-ethyl	451.2	3.15	5.58	0	5	8
Ethoxysulfuron	398.4	1.74	1.65	2	11	10
Fenoxaprop	333.7	1.95	3.74	1	6	6
Fenoxaprop-ethyl	361.8	2.42	4.12	0	6	7
Fenuron	164.2	1.67	1.12	1	3	3
Flamprop-m-isopropyl	363.8	3.10	4.58	0	4	7
Flamprop-methyl	335.8	2.64	3.82	0	4	6
Flazasulfuron	407.3	1.12	1.56	2	10	7
Fluazifop	327.3	1.86	3.57	1	5	6
Fluazifop-butyl	383.4	2.79	4.81	0	5	9
Flumetsulam	325.3	1.32	1.64	1	7	3
Flumiclorac-pentyl	423.9	2.46	3.70	0	6	9
Flumioxazin	354.3	1.31	1.08	0	6	3
Fluoroglyphen-ethyl	447.8	2.61	4.48	0	8	9
Flupropacil	390.7	2.78	2.66	0	6	4
Flupyralsulfuron-methyl	465.4	0.91	1.69	2	12	9
Fluroxypyr	255.0	0.23	1.57	3	5	5
Fluthiacet-methyl	403.9	2.79	3.31	0	6	5
Fosamine	168.1	-0.96	-0.46	4	6	6
Glufosinate	181.1	-3.76	-3.60	4	5	7
Glyphosate	169.1	-2.37	-0.77	4	6	7
Halosulfuron-methyl	434.8	0.46	0.58	2	13	9
Haloxypyr-methyl	375.7	2.34	4.12	0	5	6
Hexazinone	252.3	3.51	2.37	0	6	2
Imazamethabenz-methyl	288.3	2.23	2.81	1	5	4

Table 1. continued

Name	Molecular mass	<i>mlog P</i>	<i>alog P</i>	H-bond donors	H-bond acceptors	Rotatable bonds
Imazapic	275.3	0.95	1.87	2	6	4
Imazamox (prop)	305.3	0.45	1.15	2	7	6
Imazapyr	261.3	0.69	1.40	2	6	4
Imazaquin	311.3	1.54	2.57	2	6	4
Imazethapyr	289.3	1.20	2.27	2	6	5
Imazosulfuron	412.8	1.26	2.06	2	11	7
Ioxynil	370.9	1.47	4.31	1	2	2
Isoproturon	206.3	2.54	2.31	1	3	4
Isoxapyrifop	383.2	2.09	3.50	0	6	6
Lactofen	461.8	2.83	5.01	0	8	9
Linuron	249.1	2.27	2.42	1	4	4
MCPA	200.6	1.16	2.32	1	3	4
MCPA-thioethyl	244.7	1.76	3.04	0	2	5
MCPB	228.7	1.73	2.75	1	3	6
Mecoprop	214.7	1.45	2.86	1	3	4
Methabenzthiazuron	221.3	1.89	1.38	1	4	3
Methoprotrotyne	271.4	2.42	1.50	2	6	8
Metobenzuron	400.5	3.22	4.30	1	7	7
Metosulam	418.3	2.22	3.58	1	9	5
Metoxuron	228.7	1.45	1.38	1	4	4
Metsulfuron-methyl	381.4	1.81	1.15	2	11	8
Monalide	239.7	2.70	4.06	1	2	5
Monolinuron	214.7	1.97	1.91	1	4	4
Naproanilide	291.4	2.68	3.94	1	3	5
Naptalam	291.3	2.38	3.51	2	4	5
NC-330 ^a	431.4	0.89	0.58	2	12	8
Nicosulfuron	410.4	0.13	0.00	2	12	9
OK-8910 ^b	253.2	2.39	0.95	2	6	4
Oxasulfuron	406.4	0.85	0.20	2	10	8
Oxyfluorfen	361.7	2.74	4.92	0	5	5
Paraquat	186.3	1.59	2.01	0	2	1
Pentachlor	239.7	2.70	3.86	1	2	5
Phenmedipham	300.3	2.52	3.14	2	6	7
Piperophos	353.5	2.05	3.56	0	4	10
Primisulfuron-methyl	468.3	1.75	2.73	2	11	11
Propanil	218.1	1.88	2.56	1	2	3
Propaquizafop	443.9	1.64	4.15	0	8	10
Prosulfuron	419.4	3.16	2.94	2	9	8
Pyraflufen-ethyl	413.2	2.46	4.01	0	6	8
Pyrazosulfuron-ethyl	414.4	0.46	0.03	2	13	10
Pyribenzoxim	609.6	5.01	7.36	0	13	13
Pyridate	378.9	3.47	6.28	0	4	11
Pyriminobac-methyl	361.4	2.14	2.64	0	9	8
Pyriothiobac	326.8	2.09	3.47	1	6	6
Quinclorac	242.1	1.56	2.64	1	3	2
Quinmerac	221.6	1.56	2.59	1	3	2
Quizalofop	344.8	1.30	3.23	1	6	6
Quizalofop-P-ethyl	372.8	1.76	3.61	0	6	7
Quizalofop-P-tefuryl	428.9	1.65	3.49	0	7	8
Rimsulfuron	431.4	0.21	-0.09	2	12	9
Sethoxydim	327.5	2.30	3.61	1	4	10
Sulcotrione	328.8	0.60	1.75	1	5	3
Sulfometuron-methyl	364.4	1.09	0.47	2	9	7
Sulfosulfuron	470.5	0.75	1.11	2	13	9
Thiazafluron	240.2	1.30	2.12	1	5	3
Thidiazimin	372.4	1.97	3.32	0	6	3
Thifensulfuron-methyl	387.4	1.35	0.62	2	11	8
Tralkoxydim	343.5	2.96	6.56	1	4	6
Triasulfuron	401.8	2.05	1.87	2	10	9
Tribenuron-methyl	395.4	2.07	1.39	1	11	8
Triclopyr	256.5	1.16	2.65	1	4	4
Triflusaluron-methyl	492.4	3.01	2.61	2	12	10

^a NC-330 = methyl 5-(4,6-dimethylpyrimidin-2-ylcarbonylsulfamoyl)-1-(2-pyridyl)pyrazole-4-carboxylate.

^b OK-8910 = *N*-(2-fluorophenyl)-5-hydroxy-3-methyl-2-oxo-1-imidazolidinecarboxamide.

of which are aromatic, but the insecticides are not consistent with this pattern. The absence of carboxylic acids from the insecticide set is striking and indicative of a general lack of acidic functional groups in insecticides. In contrast, acidic groups such as carboxylic acids and acylsulfonamides are prevalent

among post-emergence herbicides. The role of weakly acidic groups in promoting phloem mobility is well known and the importance of phloem mobility in moving post-emergence herbicides to the growing points of the plant has been pointed out.³⁷ The absence of acidic groups from the ranks of insecticides

Table 2. Insecticide training set

Name	Molecular mass	mlog P	alog P	H-bond donors	H-bond acceptors	Rotatable bonds
Abamectin	905.1	3.19	4.84	3	16	13
Acephate	183.2	-0.76	0.00	1	4	4
Acetamiprid	222.7	0.56	1.98	0	4	4
Acrinathrin	541.5	3.56	6.74	0	6	11
Alanycarb	399.5	2.97	3.23	0	7	13
Aldicarb	190.3	1.60	0.74	1	4	5
Aldoxycarb	222.3	0.67	-0.58	1	6	5
Aldrin	364.9	5.36	3.55	0	0	0
Allethrin	302.4	3.40	4.05	0	3	6
Amitraz	293.4	3.97	5.32	0	3	4
Aspon	378.4	4.77	5.71	0	5	14
Azadirachtin	732.7	0.37	-1.77	3	16	13
Azamethiphos	324.7	0.38	2.39	0	7	5
Azinphos-methyl	317.3	0.65	3.97	0	6	5
Barthrin	336.8	3.40	4.31	0	4	5
Bendiocarb	223.2	1.48	1.84	1	5	3
Benfuracarb	410.5	2.94	3.32	0	7	11
Bensultap	431.6	1.76	2.74	0	5	9
Benzethazet	336.4	3.36	4.28	0	2	3
Benzoximate	363.8	2.42	4.49	0	6	8
Bifenazate	300.4	2.80	3.33	2	5	7
Bifenthrin	422.9	5.40	6.30	0	2	6
Bomyl	310.2	1.22	0.42	0	8	11
Bornate	253.4	2.19	3.01	0	3	5
Bromophos	366.0	3.22	4.80	0	3	4
Bromophos-ethyl	394.1	3.76	5.48	0	3	6
Bromopropylate	428.1	3.46	5.26	1	3	6
Buprofezin	305.4	3.50	4.28	0	4	3
Butacarb	263.4	3.58	4.78	1	3	5
Butocarboxim	190.3	1.60	1.50	1	4	5
Butonate	327.5	1.25	2.45	0	5	7
Cadusafos	270.4	1.87	4.03	0	2	8
Carbaryl	201.2	2.37	2.52	1	3	3
Carbofuran	221.3	1.74	1.74	1	4	3
Carbophenothion	342.9	2.57	5.16	0	2	8
Carbosulfan	380.6	3.74	4.88	0	5	11
Cartap	237.3	0.00	-0.26	4	5	9
Chloproxyfen	366.9	4.20	6.67	0	2	7
Chlordane	409.8	5.47	4.16	0	0	0
Chlordecone	490.6	4.91	5.36	0	1	0
Chlordimeform	196.7	2.29	2.79	0	2	2
Chlorethoxyfos	336.0	3.47	4.15	0	3	6
Chlorfenapyr	407.6	3.32	5.87	0	3	5
Chlorfenson	303.2	1.94	3.81	0	3	3
Chlorfenvinphos	359.6	3.25	4.21	0	4	7
Chlorfluazuron	540.7	4.01	6.07	2	6	7
Chlormephos	234.7	1.37	2.99	0	2	6
Chlorobenzilate	325.2	3.00	4.30	1	3	6
Chlorphoxim	332.7	2.68	4.69	0	5	8
Chlorpyrifos	350.6	3.36	4.97	0	4	6
Chlorpyrifos-methyl	322.5	2.79	4.28	0	4	4
Cinerin 1	316.4	3.63	4.40	0	3	6
Cinerin II	360.5	2.98	3.66	0	5	8
Cloethocarb	259.7	1.76	2.14	1	5	7
Clofentezine	303.2	4.00	4.66	0	4	2
CPMC	185.6	1.67	2.04	1	3	3
Crotoxyphos	314.3	2.17	2.51	0	6	8
Cyanofenphos	303.3	2.20	4.45	0	3	6
Cyanophos	243.2	1.59	3.01	0	4	5
Cycloprothrin	482.4	3.25	6.58	0	5	10
Cyfluthrin	434.3	3.48	5.94	0	4	8
Cyhalothrin	449.9	3.68	5.65	0	4	8
Cypermethrin	416.3	3.37	5.80	0	4	8
Cyphenophrin	375.5	3.37	5.61	0	4	8
Cyromazine	166.2	1.63	0.03	5	6	4
DDT	354.5	5.24	6.39	0	0	2
Deltamethrin	505.2	3.58	6.40	0	4	8
Demeton-S-methyl	230.3	0.54	1.52	0	3	7
Diafenthiuron	384.6	4.12	7.07	2	3	8
Dialifor	393.9	1.57	4.17	0	5	8
Diazinon	304.4	3.38	4.51	0	5	7
Dibrom	380.8	2.36	2.57	0	4	5
Dicapthon	297.7	2.30	3.44	0	6	5
Dichlofenthion	315.2	3.36	4.69	0	3	6

Table 2. continued

Name	Molecular mass	mlog P	alog P	H-bond donors	H-bond acceptors	Rotatable bonds
Dichlorvos	221.0	1.25	1.84	0	4	4
Dicofol	370.5	4.22	5.66	1	1	3
Dicrotophos	237.2	0.68	-0.08	0	6	6
Dicyclanil	190.2	0.61	0.43	5	6	5
Dieldrin	380.9	4.45	2.67	0	1	0
Dienochlor	474.6	5.63	1.19	0	0	1
Diflubenzuron	310.7	2.86	3.33	2	4	5
Dimefox	154.1	-0.04	-0.10	0	3	2
Dimethoate	229.3	-0.35	0.56	1	4	6
Dimetilan	240.3	2.09	0.48	0	7	5
Dinocap	364.4	3.66	5.86	0	8	11
Dioxabenzofos	216.2	2.16	2.70	0	3	1
Dioxacarb	223.2	1.48	1.26	1	5	4
Disulfoton	274.4	2.06	2.94	0	2	9
DMCP	252.7	0.86	3.10	0	3	4
DNOC	198.1	0.76	2.14	1	7	3
Endosulfan	406.9	3.52	2.12	0	3	0
Endrin	380.9	4.45	2.67	0	1	0
EPBP	347.2	3.36	5.45	0	2	5
EPN	323.3	2.60	4.37	0	5	6
Ethiofencarb	225.3	2.26	2.05	1	3	6
Ethion	384.5	1.44	4.82	0	4	12
Ethoprophos	242.3	1.25	3.21	0	2	8
Ethylan	307.3	5.47	6.40	0	0	5
Etofenprox	376.5	3.57	6.25	0	3	9
Etozazole	359.4	3.91	5.86	0	3	5
Etrifos	292.3	2.85	3.88	0	6	7
Fenamiphos	303.4	2.82	3.24	1	4	7
Fenchlorphos	321.5	3.07	4.52	0	3	4
Fenitrothion	277.2	2.30	3.39	0	6	5
Fenobucarb	207.3	2.54	3.11	1	3	5
Fenoxycarb	301.3	1.98	3.23	1	5	9
Fenpropathrin	349.4	3.02	5.36	0	4	7
Fenpyroximate	421.5	3.41	5.02	0	7	9
Fensulfothion	308.4	2.18	2.24	0	4	7
Fenthion	253.4	2.00	2.55	0	3	8
Fenthion	278.3	2.78	3.53	0	3	5
Fenvalerate	419.9	3.64	6.80	0	4	9
Fipronil	437.2	1.98	5.23	2	5	4
Flubrocyrthrin	530.4	3.35	7.67	0	5	11
Flucyclohexuron	483.9	4.24	5.79	2	6	10
Flucythrinate	451.5	3.05	6.88	0	5	11
Flufenoxuron	488.8	3.86	5.78	2	5	7
Flufenprox	450.9	3.88	6.30	0	3	9
Flufenzine	304.7	4.00	4.42	0	4	2
Fonofos	246.3	2.06	3.17	0	1	5
Formetanate	221.3	1.39	1.28	1	5	5
Formothion	257.3	-0.45	0.35	0	5	7
Fosthiazate	283.4	1.05	2.60	0	4	6
Fosthietan	241.3	0.85	2.91	0	4	5
Furathiocarb	382.5	3.29	4.05	0	7	10
Halfenprox	477.3	3.88	8.43	0	3	9
Halofenozide	330.8	3.16	4.01	1	4	6
Heptachlor	373.3	5.12	3.67	0	0	0
Heptenophos	250.6	2.07	0.70	0	4	4
Hexythiazox	352.9	3.98	4.34	1	4	4
Hydroprene	266.4	4.33	5.16	0	2	10
Imidacloprid	255.7	1.84	2.29	1	7	3
Indoxacarb	527.8	3.74	5.64	0	10	8
Iodofenphos	413.0	3.36	5.26	0	3	4
IPSP	300.4	2.57	4.59	0	2	8
Isazofos	313.7	3.75	3.85	0	6	7
Isobenzan	411.8	4.21	3.57	0	1	0
Isofenphos	345.4	3.28	3.9	1	5	9
Isoprocarb	193.2	2.26	2.71	1	3	4
Isoprothiolane	290.4	1.83	1.99	0	4	6
Isothioate	260.4	1.73	2.67	0	2	7
Isoxathion	313.3	3.49	4.48	0	5	7
Jasmolin II	374.5	3.20	4.06	0	5	9
Kinoprene	276.4	4.48	5.09	0	2	11
Leptophos	412.1	3.48	5.90	0	2	4
Lindane	290.8	4.09	4.65	0	0	0
Malathion	330.4	0.90	1.84	0	6	11
Mecarbam	329.4	1.30	2.21	0	6	11

Table 2. continued

Name	Molecular mass	mlog P	alog P	H-bond donors	H-bond acceptors	Rotatable bonds
Menazon	281.3	0.78	1.64	4	7	7
Mephosfolan	269.3	1.54	2.96	0	4	5
Metepa	215.2	1.97	0.05	0	4	3
Methamidophos	141.1	-1.22	0.00	2	3	3
Methidathion	302.3	0.01	3.2	0	6	6
Methiocarb	225.3	2.26	2.55	1	3	4
Methomyl	162.2	0.87	0.62	1	4	4
Methoprene	310.5	3.97	4.47	0	3	11
Methoxychlor	345.7	3.34	4.85	0	2	4
Methoxyfenozide	368.5	3.05	4.64	1	5	7
Metolcarb	165.2	1.67	1.99	1	3	3
Mevinphos	224.2	0.76	0.32	0	6	6
Mexacarbate	222.3	1.74	2.72	1	4	4
Mirex	545.5	6.52	6.41	0	0	0
Monocrotophos	223.2	0.35	-0.33	1	6	6
Morphothion	285.3	-0.47	0.46	0	5	6
NC-170 ^a	417.1	2.12	2.84	0	5	4
Nicotine	162.2	1.27	1.02	0	2	1
Nitenpyram	270.7	1.55	2.12	1	6	6
Omethoate	213.2	-1.14	-0.18	1	5	6
Oxydemeton-methyl	246.3	-0.38	-0.12	0	4	7
Oxydeprofos	260.3	-0.03	0.30	0	4	7
Oxydisulfoton	290.4	1.12	1.30	0	3	9
Paraoxon	275.2	1.84	2.87	0	7	7
Parathion	291.3	2.60	3.61	0	6	7
Parathion-methyl	263.2	1.98	2.92	0	6	5
Permethrin	391.3	4.04	5.73	0	3	7
Phenothrin	350.5	4.04	5.54	0	3	7
Phenthoate	320.4	1.65	3.52	0	4	8
Phorate	260.4	1.73	3.30	0	2	8
Phosalone	367.8	1.95	4.43	0	5	7
Phosfolan	255.3	1.21	2.54	0	4	5
Phosmet	317.3	0.50	2.81	0	5	5
Phosphamidon	299.7	1.60	0.38	0	6	8
Phoxim	298.3	2.40	4.17	0	5	8
Pirimicarb	238.3	2.33	1.92	0	6	4
Pirimiphos-ethyl	333.4	3.70	4.02	0	6	9
Pirimiphos-methyl	305.3	3.14	3.34	0	6	7
Prallethrin	300.4	3.40	3.58	0	3	6
Profenofos	373.6	2.14	5.04	0	3	7
Promecarb	207.3	2.54	3.18	1	3	4
Propafos	304.3	2.82	3.95	0	4	9
Propargite	350.5	3.40	4.81	0	5	8
Propetamphos	281.3	2.48	1.51	1	5	8
Propoxur	209.2	1.45	2.02	1	4	5
Prothiofos	345.2	2.84	5.50	0	2	7
Prothoate	285.4	1.03	2.00	1	4	9
Pymetrozine	217.2	1.06	1.09	0	6	2
Pyraclofos	360.8	1.57	4.28	0	5	8
Pyrethrin I	328.5	3.77	4.58	0	3	7
Pyridaben	364.9	3.95	3.45	0	3	5
Pyridaphenthion	340.3	3.31	4.07	0	6	7
Pyriproxyfen	321.4	2.50	4.64	0	4	7
Quinalphos	298.3	2.40	3.07	0	5	6
Quinomethionate	234.3	0.18	3.07	0	3	0
Resmethrin	338.5	3.91	4.85	0	3	7
Rotenone	394.4	0.86	2.60	0	6	3
Schradan	286.3	2.05	-0.62	0	7	6
Silafluofen	408.6	4.62	6.31	0	2	9
Spinosyn D	746.0	3.48	4.82	0	11	9
Sulfluramid	527.2	3.54	6.04	1	3	9
Sulfotep	322.3	3.58	3.84	0	5	10
Sulprofos	322.4	2.57	4.56	0	2	8
Tau-fluvalinate	502.9	3.35	7.28	1	5	10
TDE	320.0	5.01	5.70	0	0	3
Tebufofenozide	352.5	3.83	5.29	1	4	7
Tebufofenpyrad	333.9	2.92	4.22	1	4	6
Tebupirimfos	318.4	2.84	4.71	0	5	7
Teflubenzuron	381.1	3.34	4.13	2	4	5
Tefluthrin	418.7	4.80	5.17	0	2	5
TEPA	173.2	0.96	-1.19	0	4	3
TEPP	290.2	2.05	2.37	0	7	10
Terbufos	288.4	2.38	3.80	0	2	8
Tetrachlorvinphos	366.0	3.00	4.05	0	4	5

Table 2. continued

Name	Molecular mass	mlog P	alog P	H-bond donors	H-bond acceptors	Rotatable bonds
Tetradifon	356.1	3.28	4.95	0	2	2
Tetramethrin	331.4	3.43	2.95	0	5	5
Thanite ^b	253.4	2.19	3.01	0	3	5
Thiocarboxime	201.2	0.72	0.68	1	5	7
Thiodicarb	354.5	2.55	2.51	0	8	10
Thiofanox	218.3	2.25	2.82	1	4	6
Thiometon	246.4	1.37	2.25	0	2	7
Thionazin	236.3	-0.15	1.39	0	4	4
Toxaphene	136.2	4.29	2.84	0	0	0
Tralomeethrin	665.0	4.26	6.57	0	4	8
Triamiphos	294.3	3.31	1.18	2	7	5
Triazamate	314.4	2.94	3.03	0	7	8
Triazophos	313.3	4.07	4.21	0	6	7
Trichlorfon	257.4	0.08	1.30	1	4	4
Trichloronate	314.6	3.10	5.15	0	2	5
Triflumuron	358.7	2.43	4.74	2	5	6
Trimethacarb	193.2	2.26	2.92	1	3	3
Vamidothion	287.3	-0.07	0.64	1	5	9
Vaniliprole	555.3	2.76	8.39	1	6	7
XMC	179.2	1.97	2.46	1	3	3
XR-100 ^c	428.4	4.24	6.19	0	4	6
Xylylcarb	179.2	1.97	2.46	1	3	3

^a NC-170 = 4-chloro-5-[(6-chloro-3-pyridinyl)methoxy]-2-(3,4-dichlorophenyl)-3(2*H*)-pyridazinone.

^b thanite = isobornyl thiocyanacetate.

^c XR-100 = 8-fluoro-4-[2-[4-[4-(trifluoromethyl)phenoxy]phenyl]ethoxy]quinazoline.

suggests that attempts to engineer plant phloem mobility into insecticides by introducing acidic functional groups may be doomed to failure. The few known phloem-mobile insecticides, eg oxamyl, acephate, apparently gain this property by virtue of other characteristics such as low log *P*.⁴⁰ Many structural units such as benzene rings, amides and esters are comparably prevalent in pharmaceuticals, herbicides and insecticides.

Both the herbicide and insecticide training sets are significantly smaller groups of compounds than

Lipinski *et al*¹¹ used in his analysis, but are comparable to some of the therapeutic groups examined by Ghose *et al*.¹² Furthermore they are strongly biased by certain major structural and mode-of-action classes, eg sulfonyl ureas (24% of the herbicide set) and organophosphates (42% of the insecticide set). Given these limitations, it seems appropriate to be less restrictive in setting parameter limits, and to exclude only extreme outliers. Based on Tables 4 and 5, modifications of the Rule of 5 for post-emergence herbicides and insecticides are proposed in Table 6. The upper limiting values in Table 4 are set at convenient values close to

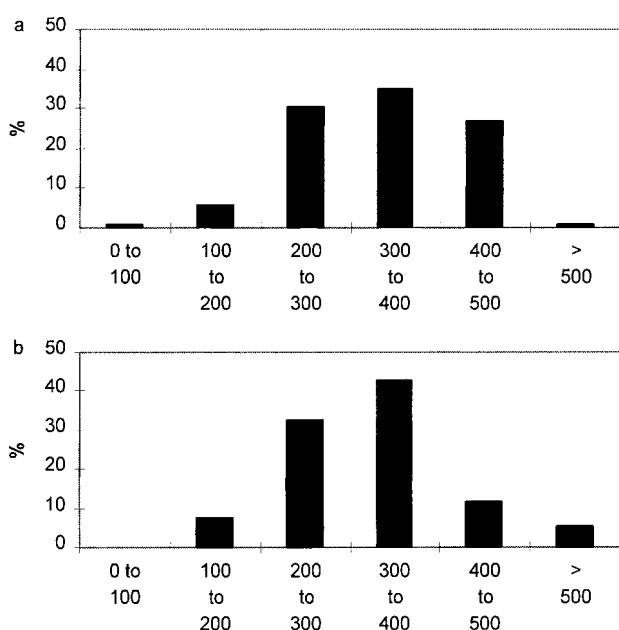


Figure 1. Molecular mass distributions for (a) post-emergence herbicides and (b) insecticides.

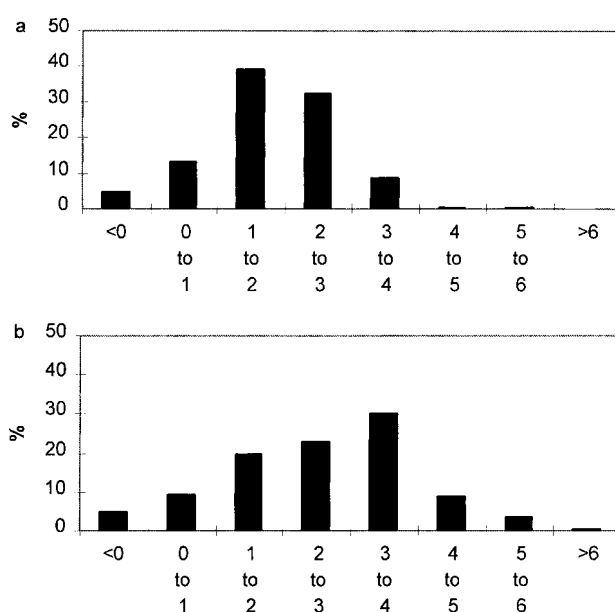


Figure 2. mlog *P* distributions for (a) post-emergence herbicides and (b) insecticides.

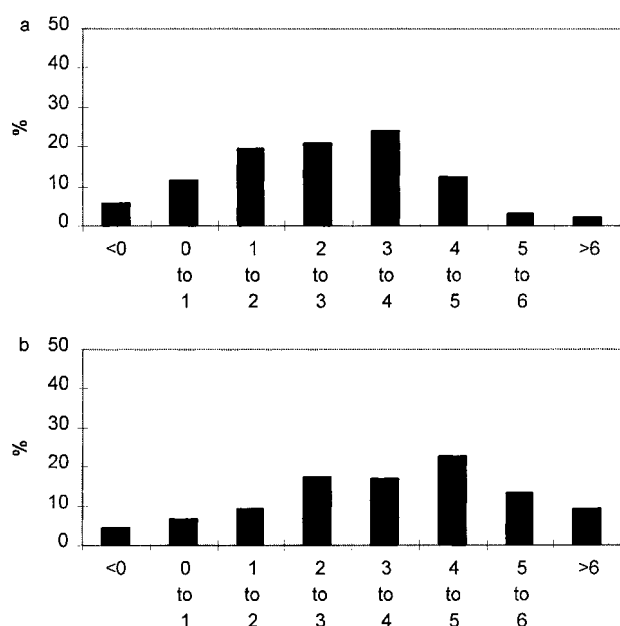


Figure 3. $\log P$ distributions for (a) post-emergence herbicides and (b) insecticides.

the 95th percentile. In certain cases, where it seemed appropriate, lower limiting values are set around the 5th percentile values shown in Tables 4 and 5. The $m\log P$ and $\log P$ criteria should be considered as alternatives. The most significant change to the Rule of 5 is the more stringent limitations on numbers of hydrogen-bond donors for both agrochemical classes. Applying the molecular mass, $m\log P$, hydrogen-bond donor, hydrogen-bond acceptor and rotatable-bond criteria from Table 6 to the appropriate compound sets results in inclusion of 88% of the herbicide and 83% of the insecticide training sets. Our experience of

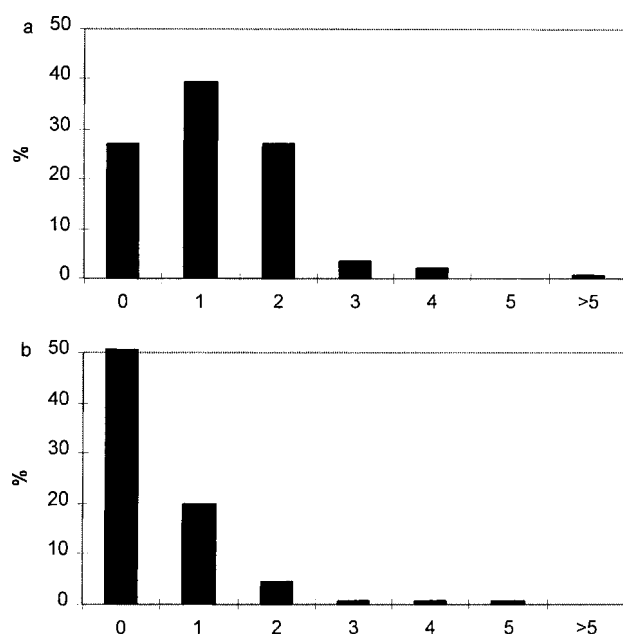


Figure 4. Distribution of hydrogen-bond donors for (a) post-emergence herbicides and (b) insecticides.

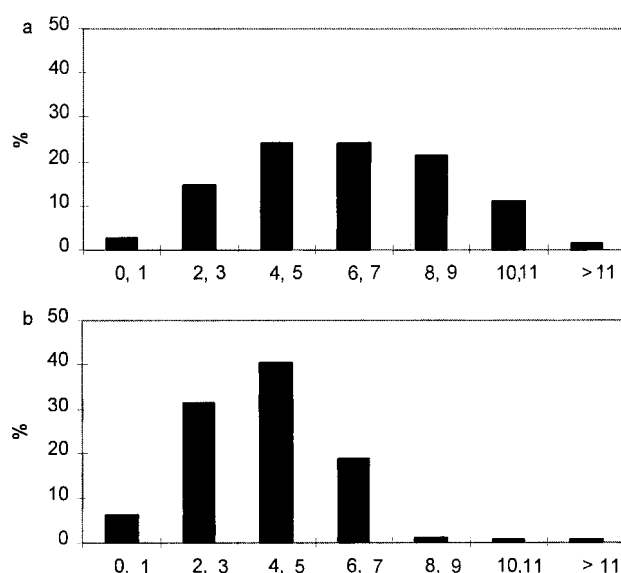


Figure 5. Distribution of hydrogen-bond acceptors for (a) post-emergence herbicides and (b) insecticides.

applying these rules to library design and selection of vendor compounds is that the lower limits are rarely violated by the compounds proposed or offered, but enforcing the upper limits has considerable impact on which compounds are selected for synthesis or purchase. Re-analysis of the herbicide data set after removal of the sulfonyl ureas ($n = 109$) gave very similar results to the full set except that 95% of the compounds have ≤ 9 hydrogen-bond acceptors rather than ≤ 12 . Re-analysis of the insecticide set after removal of the prolific organophosphate class ($n = 141$) gave similar results to the full set and the only significant change to the rules proposed in Table 6 would be an upward shift of $c 0.5$ in the lower limits for both $m\log P$ and $\log P$.

Tables 7 and 8 show the parameter values for several

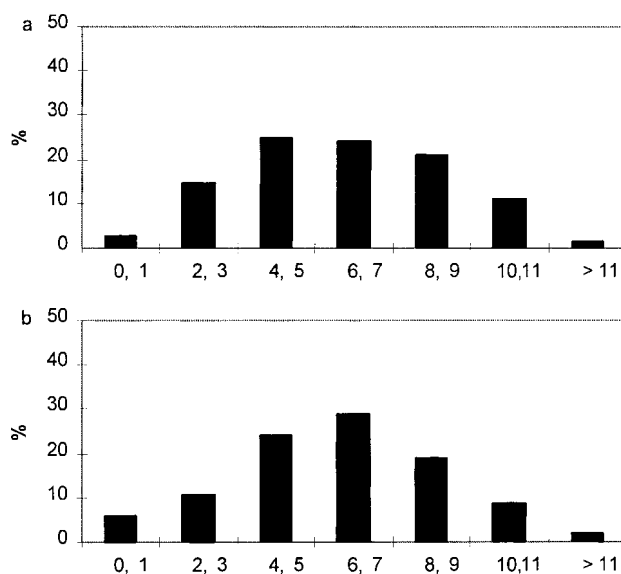


Figure 6. Distribution of rotatable bonds for (a) post-emergence herbicides and (b) insecticides.

Table 3. Functional group distributions

Functional group	CMC ^a (n = 6454) (%)	Herbicides (n = 136) (%)	Insecticides (n = 243) (%)
Carboxyl	15.1	22.8	0.0
Alcohol	25.8	0.7	2.4
Aldehyde	0.5	0.0	0.0
Aliphatic primary amine	5.7	1.5	0.0
Aliphatic secondary amine	9.1	0.7	0.0
Aliphatic tertiary amine	29.6	0.0	0.0
Aromatic primary amine	5.4	2.2	2.0
Aromatic secondary amine	7.2	0.7	0.8
Aromatic tertiary amine	10.3	0.7	0.0
Carboxamide	27.1	18.2	11.2
Nitro	2.6	4.4	4.0
Pyridine	8.1	16.8	4.4
Pyrimidine	2.4	16.1	3.2
Pyrrole	4.4	0.0	0.4
Benzene	70.3	73.7	55.2
Furan	2.0	0.0	0.4
Thiophene	1.9	0.7	0.0
Imidazole	6.0	1.5	0.0
Ester	18.2	26.3	20.8
Sulfonamide	4.5	21.9	0.4
Heterocyclic (any)	66.8	62.8	29.6
Heterocyclic (aromatic)	28.4	54.7	17.2

^a Comprehensive Medicinal Chemistry Database. Data from Reference 12.

Table 4. Summary of property characteristics of post-emergence herbicides^a

Parameter	Figure	Minimum	5th percentile	Average	95th percentile	Maximum
Molecular mass	1a	84	184	329	461	610
mlog <i>P</i>	2a	−3.8	−0.2	1.7	3.5	5.0
alog <i>P</i>	3a	−4.7	−0.4	2.5	4.9	7.4
H-bond donor	4a	0	0	1.2	3	6
H-bond acceptor	5a	2	2	6.4	12	15
Rotatable bonds	6a	0	2	6.1	10	13

^a *n* = 136.

Table 5. Summary of property characteristics of insecticides^a

Parameter	Figure	Minimum	5th percentile	Average	95th percentile	Maximum
Molecular mass	1b	136	186	324	491	905
mlog <i>P</i>	2b	−1.2	0.0	2.6	4.6	6.5
alog <i>P</i>	3b	−1.8	0.0	3.5	6.4	8.4
H-bond donor	4b	0	0	0.4	2	5
H-bond acceptor	5b	0	1	4.1	7	16
Rotatable bonds	6b	0	0	6.1	11	14

^a *n* = 243.

recently introduced post-emergence herbicides and insecticides. These compounds were all members of the training sets. Values outside the ranges proposed in Table 6 are indicated. Overall, these newer compounds conform well to the rules proposed above and their parameter values are quite close to the averages in Tables 4 and 5. There is an apparent trend towards higher molecular mass in the new herbicide set. Of the herbicides, only sulfosulfuron has an acidic group that would impart phloem mobility, although pyriminobac-methyl is presumably readily hydrolyzed *in vivo* to its parent carboxylic acid. On the basis of their

structures, three of the herbicides are presumably protoporphyrinogen IX oxidase (protox) inhibitors (carfentrazone-ethyl, fluthiacet-methyl, pyraflufen-ethyl) and two are acetolactate synthase inhibitors (pyriminobac-methyl, sulfosulfuron). Protox inhibition causes rapid damage to treated tissue and is thus often considered to be incompatible with phloem mobility. Among the newer insecticides there are weak trends towards higher molecular mass and an increased number of hydrogen-bond acceptors. The newer insecticides include three neurotoxins (imidacloprid, fipronil and indoxacarb), an uncoupler of

Table 6. Proposed rules for insecticides and post-emergence herbicides

Parameter	Pharmaceuticals ^a	Post-emergence herbicides	Insecticides
Molecular mass	≤ 500	≥ 150 and ≤ 500	≥ 150 and ≤ 500
mlog <i>P</i>	≤ 4.15	≤ 3.5	≥ 0 and ≤ 5.0
alog <i>P</i>	—	≤ 5.0	≥ 0 and ≤ 6.5
H-bond donors	≤ 5	≤ 3	≤ 2
H-bond acceptors	≤ 10	≥ 2 and ≤ 12	≥ 1 and ≤ 8
Rotatable bonds	—	≤ 12	≤ 12

^a See Reference.¹¹.**Table 7.** Parameter values of recently introduced post-emergence herbicides

Parameter	Carfentrazone-ethyl	Fluthiacet-methyl	Pyraflufen-ethyl	Pyriminobac-methyl	Sulfosulfuron
Molecular mass	412.2	403.9	413.2	361.4	364.4
mlog <i>P</i>	4.08 ^a	2.79	2.46	2.14	1.09
alog <i>P</i>	4.18	3.31	4.00	2.64	0.47
H-bond donor	0	0	0	0	2
H-bond acceptor	6	6	6	9	9
Rotatable bonds	7	5	8	8	7

^a Value outside the range proposed in Table 6.

Parameter	Imidacloprid	Fipronil	Indoxacarb	Chlorfenapyr	Methoxyfenozide
Molecular mass	255.7	437.2	527.8 ^a	407.6	368.5
mlog <i>P</i>	1.84	1.98	3.74	3.32	3.04
alog <i>P</i>	2.29	5.23	5.65	5.87	4.64
H-bond donor	1	2	0	0	1
H-bond acceptor	7	5	10 ^a	5	5
Rotatable bonds	3	4	8	5	7

Table 8. Parameter values of recently introduced insecticides^a Outside the range proposed in Table 6.

oxidative phosphorylation (chlorfenapyr) and an ecdysone agonist (methoxyfenozide).

The major argument against the application of criteria such as those proposed in Table 6 to compound selection is that it eliminates the chance of discovering a radically new type of lead with unusual properties. This is undeniably true: application of the rules is a conservative strategy to reduce the risk of failure by staying within the boundaries of what is already known to work. It should be pointed out that the number of organic compounds within the boundaries of the rules remains infinite for all practical purposes.

Glyphosate has the following parameters: molecular mass = 169, mlog *P* = −2.37, alog *P* = −0.77, H-bond donors = 4, H-bond acceptors = 6, rotatable bonds = 7. The four hydrogen bond donors place it outside the limits set in Table 6. Given glyphosate's enormous economic significance, this is a troubling result. Interestingly, it has been proposed that uptake of glyphosate into the leaf occurs by a hydrophilic pathway.⁵⁹ This may differ from the way in which the majority of herbicides penetrate the leaf cuticle. A different set of criteria could possibly be developed for compounds that follow this hydrophilic pathway. In addition, it has been proposed that active transport

may play a role in glyphosate uptake.⁶⁰ Lipinski *et al* have pointed out that actively transported compounds are not subject to the Rule of 5.¹¹ It is also worth noting that glyphosate is an unusual compound, in that virtually every structural change explored abolishes herbicidal activity.⁶¹ Thus discovery of glyphosate would be unlikely in the course of a typical lead discovery-lead optimization program. Among the outliers in the insecticide data set are complex natural products, including azadirachtin and abamectin.

4 CONCLUSION

A set of simple criteria have been derived that can guide screening-compound purchase and library synthesis towards compounds that are more likely to have whole-organism insecticidal or post-emergence herbicidal activity. Differences have been noted between the optimal parameter ranges for post-emergence herbicides and for insecticides. Lead compounds with whole-organism activity are generally regarded as a more favorable starting point for lead optimization than compounds with only *in vitro* activity.

REFERENCES

- 1 Terrett NK, *Combinatorial Chemistry*, Oxford Chemistry Masters, Vol 2, Oxford University Press, Oxford (1998).
- 2 *Combinatorial Chemistry and Molecular Diversity in Drug Discovery*, ed by Gordon EM and Kerwin JF, Wiley-Liss, New York, NY, USA (1998).
- 3 Bunin BA, *The Combinatorial Index*, Academic Press, San Diego (1998).
- 4 *Combinatorial Chemistry: Synthesis and Application*, ed by Wilson SR and Czarnik AW, John Wiley and Sons, New York, NY, USA (1997).
- 5 Van Drie JH and Lajiness MS, Approaches to virtual library design. *Drug Discovery Today* 3:274–283 (1998).
- 6 Agrafiotis DK, Myslik JC and Salemme FR, Advances in diversity profiling and combinatorial series design. *Mol Diversity* 4:1–22 (1999).
- 7 Matter H, Selecting optimally diverse compounds from structure databases: a validation study of two-dimensional and three-dimensional molecular descriptors. *J Med Chem* 40:1219–1229 (1997).
- 8 Patterson DE, Cramer RD, Ferguson AM, Clark RD and Weinberger LE, Neighborhood behavior: a useful concept for validation of molecular diversity descriptors. *J Med Chem* 39:3049–3059 (1996).
- 9 Cramer RD, Clark RD, Patterson DE and Ferguson AM, Bioisosterism as a molecular diversity descriptor: steric fields of single 'topomeric' conformers. *J Med Chem* 39:3060–3069 (1996).
- 10 Fecik RA, Frank KE, Gentry EJ, Menon SR, Mitscher LA and Telikepalli H, The search for orally active medications through combinatorial chemistry. *Med Res Rev* 18:149–185 (1998).
- 11 Lipinski CA, Lombardo F, Dominy BW and Feeney PJ, Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Adv Drug Delivery Rev* 23:3–25 (1997).
- 12 Ghose AK, Viswanadhan VN and Wendoloski JJ, A knowledge-based approach in designing combinatorial or medicinal chemistry libraries for drug discovery. 1. A qualitative and quantitative characterization of known drug databases. *J Comb Chem* 1:55–68 (1999).
- 13 Pagliara A, Reist M, Geinoz S, Carrupt PA and Testa B, Evaluation and prediction of drug permeation. *J Pharm Pharmacol* 51:1339–1357 (1999).
- 14 Van de Waterbeemd H, Camenisch G, Folkers G, Chretien JR and Raevsky OA, Estimation of blood-brain barrier crossing of drugs using molecular size and shape, and H-bonding descriptors. *J Drug Targeting* 6:151–165 (1998).
- 15 Clark DE, Rapid calculation of polar molecular surface area and its application to the prediction of transport phenomena. 2. Prediction of blood brain barrier penetration. *J Pharm Sci* 88:815–821 (1999).
- 16 Abraham MH, Chadha HS, Martins F, Mitchell RC, Bradbury MW and Gratton JA, Hydrogen bonding part 46: A review of the correlation and prediction of transport properties by an LFER method: physicochemical properties, brain penetration and skin permeability. *Pestic Sci* 55:78–88 (1999).
- 17 Kramer SD, Absorption prediction from physicochemical parameters. *Pharm Sci Technol Today* 2:373–380 (1999).
- 18 Norinder U, Osterburg T and Artursson P, Theoretical calculation and prediction of intestinal absorption of drugs in humans using MolSurf parameterization and PLS statistics. *Eur J Pharm Sci* 8:49–56 (1999).
- 19 Clark DE, Rapid calculation of polar molecular surface area and its application to the prediction of transport phenomena. 1. Prediction of intestinal absorption. *J Pharm Sci* 88:807–814 (1999).
- 20 Palm K, Luthman K, Ungell AL, Stranlund G, Beigi F, Lundahl P and Artursson P, Evaluation of dynamic polar molecular surface area as predictor of drug absorption: comparison with other computational and experimental predictors. *J Med Chem* 41:5382–5392 (1998).
- 21 Winiwarter S, Bonham NM, Ax F, Hallberg A, Lennernas H and Karlen A, Correlation of human jejunal permeability (*in vivo*) of drugs with experimentally and theoretically derived parameters. A multivariate data analysis approach. *J Med Chem* 41:4939–4949 (1998).
- 22 Walters WP, Ajay and Murcko MA, Recognizing molecules with drug-like properties. *Curr Opin Chem Biol* 3:384–387 (1999).
- 23 Ajay, Walters WP and Murcko MA, Can we learn to distinguish between 'drug-like' and 'nondrug-like' molecules? *J Med Chem* 41:3314–3324 (1998).
- 24 Sadowski J and Kubinyi H, A scoring scheme for discriminating between drugs and nondrugs. *J Med Chem* 41:3325–3329 (1998).
- 25 Leo AJ, Calculating log P_{oct} from structures. *Chem Rev* 1993, 93:1281–1304.
- 26 Moriguchi I, Hirono S, Liu Q, Nakagome I and Matsushita Y, A simple method of calculating the octanol water partition-coefficient. *Chem Pharm Bull* 40:127–130 (1992).
- 27 Stenberg P, Luthman K, Ellens H, Lee CP, Smith PL, Lago A, Elliott JD and Artursson P, Prediction of the intestinal absorption of endothelin receptor antagonists using three theoretical methods of increasing complexity. *Pharmaceut Res* 16:1520–1526 (1999).
- 28 Ghose AK and Crippen GM, Atomic physicochemical parameters for three dimensional structure-directed quantitative structure-activity relationships. I. Partition coefficients as a measure of hydrophobicity. *J Comput Chem* 7:565–577 (1986).
- 29 Ghose AK, Pritchett A and Crippen GM, Atomic physicochemical parameters for three dimensional structure directed quantitative structure-activity relationships. III. Modeling hydrophobic interactions. *J Comput Chem* 9:80–90 (1988).
- 30 Viswanadhan VN, Ghose AK, Revankar GR and Robins RK, Atomic physicochemical parameters for 3-dimensional structure directed quantitative structure-activity relationships. IV. Additional parameters for hydrophobic and dispersive interactions and their application for an automated superposition of certain naturally-occurring nucleoside antibiotics. *J Chem Inf Comput Sci* 29:163–172 (1989).
- 31 Baur P, Mechanistic aspects of foliar penetration of agrochemicals and the effect of adjuvants. *Rec Res Dev Agric Food Chem* 2:809–837 (1998).
- 32 Foy CL, Adjuvants—current technology and trends, in *Pesticide Formulation Adjuvant Technology*, Formulations Forum 94 ed by Foy CL and Pritchard DW, CRC Press, Boca Raton, FL, USA, pp 323–352 (1996).
- 33 Nalewaja JD, Behavior, applicability and efficacy of non-surfactant adjuvants. *FRI Bull* 193:186–192 (1996).
- 34 Steurbaut W, Adjuvants for use with foliar fungicides. *Pestic Sci* 38:85–91 (1993).
- 35 Kirkwood RC, Recent developments in our understanding of the plant cuticle as a barrier to the foliar uptake of pesticides. *Pestic Sci* 55:69–77 (1999).
- 36 Baur P and Schoenherr J, Die Aufnahme systemischer Wirkstoffe über Blätter: Grundlagen und Optimierung. *Gartenbauwissenschaft* 61:105–115 (1996).
- 37 Kleier DA, Grayson BT and Hsu FC, The phloem mobility of pesticides. *Pestic Outlook* 9:26–30 (1998).
- 38 Hsu FC and Kleier DA, Phloem mobility of xenobiotics. VIII. A short review. *J Exp Bot* 47:1265–1271 (1996).
- 39 Kleier DA and Hsu FC, Phloem mobility of xenobiotics. VII. The design of phloem systemic pesticides, *Weed Sci* 44:749–756 (1996).
- 40 Brudenell AJP, Baker DA and Grayson BT, Phloem mobility of xenobiotics: tabular review of physicochemical properties governing the output of the Kleier model. *Plant Growth Regul* 16:215–231 (1995).
- 41 Kleier DA, Phloem mobility of xenobiotics. V. Structural requirements for phloem systemic pesticides. *Pestic Sci* 42:1–11 (1994).
- 42 Rigitano RLO, Bromilow RH, Briggs GG and Chamberlain K,

- Phloem translocation of weak acids in *Ricinus communis*. *Pestic Sci* **19**:113–133 (1987).
- 43 Cronin MTD, Dearden JC, Moss GP and Murray-Dickson G, Investigation of the mechanism of flux across human skin in vitro by quantitative structure-permeability relationships. *Eur J Pharm Sci* **7**:325–330 (1999).
 - 44 Briggs GG, Bromilow RH and Evans AA, Relationship between lipophilicity and root uptake and translocation of non-ionized chemicals by barley. *Pestic Sci* **13**:495–504 (1982).
 - 45 Sicbaldi F, Sacchi GA, Trevisan M and Del Re AAM, Root uptake and xylem translocation of pesticides from different chemical classes. *Pestic Sci* **50**:111–119 (1997).
 - 46 Lankford WT, factors affecting penetration through the cuticles of insects and mites, in *Ernst Schering Research Foundation Workshop 12: Interactions between Adjuvants, Agrochemicals and Target Organisms*, pp 107–27 (1994).
 - 47 Coats JR, Williams JW, Chang C, Lee AH and Metcalf RL, Structure-activity for uptake and toxicity of DDT-type insecticides utilizing an NMR method for estimating σ^* . *Environ Toxicol Chem* **8**:45–52 (1989).
 - 48 De Licastro SA, Zerba EN and Casabe N, The relation between viscosity and penetration of some diethyl *p*-substituted phenyl phosphorothionates and oil carriers into the cuticle of *Triatoma infestans*. *Pestic Biochem Physiol* **19**:53–59 (1983).
 - 49 Lewis CT, The Penetration of cuticle by insecticides, in *Cuticle Tech. Arthropods*, ed by Miller TA, Springer, New York, Secaucus, NJ, USA. pp 367–400 (1980).
 - 50 *Farm Chemicals Handbook '97*, Meister Publishing, Willoughby, Ohio, USA (1997).
 - 51 Hopkins WL, *Global Herbicide Directory*, 1st edn, Ag Chem Information Services, Indianapolis, Indiana, USA (1994).
 - 52 Hopkins WL, *Global Herbicide Directory*, 2nd edn, Ag Chem Information Services, Indianapolis, Indiana, USA (1997).
 - 53 Bryant R and Bite M, Ag Chem. New Compound Review, Vol 17, Agranova, Orpington, UK (1999).
 - 54 Tripos Inc, 1699 South Hanley Road, St Louis, MO 63144, U S A.
 - 55 Based on an SPL script written by Dr James Blake (Pfizer, USA substantially modified by Dr Ted Fujimoto (Rohm and Haas Company, USA).
 - 56 Molecular Simulations Inc, 9685 Scranton Road, San Diego, CA 92121, U S A.
 - 57 Buchwald P and Bodor N, Octanol-water partition: searching for predictive models. *Curr Med Chem* **5**:353–380 (1998).
 - 58 Aizawa H, *Metabolic Maps of Pesticides*, Vol 2, Academic Press, San Diego, CA, U S A (1989).
 - 59 Franz JE, Mao MK and Sikorski JA, Uptake, transport and metabolism of glyphosate in plants, in *Glyphosate: A Unique Global Herbicide*, American Chemical Society, Washington DC, U S A, pp 143–186 (1997).
 - 60 Mervosh TL and Balke NE, Effects of calcium, magnesium and phosphate on glyphosate absorption by cultured plant cells. *Weed Sci* **39**:347–353 (1991).
 - 61 Franz JE, Mao MK and Sikorski JA, Structure-activity relationships for glyphosate and its derivatives, in *Glyphosate: A Unique Global Herbicide*, American Chemical Society, Washington DC, U S A, pp 441–520 (1997).